Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *AFRIDOC_MT* | ***Context:*** *Amhara Region Agricultural & Microfinance Document Translation — Cooperative and Rural MFI End-Users*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is text-only translation [Q8, Q44, Q201], matching the deployment's text-only requirement. Document-level outputs are produced by realigning sentence-level or pseudo-document translations into full documents [Q51, Q57], partially preserving paragraph structure via maintained empty rows [Q119]. However, no metric assesses structural fidelity (preservation of numbered clauses, tables, signature blocks) [regional context: structure_preservation_gap]. The benchmark also reports specific output-form pathologies for African-language translations: under-generation, over-generation, repetition of words and phrases, off-target translations [Q6], and inconsistent diacritics [Q77] — these failure modes would be especially harmful in a legal-administrative deployment where exact term repetition matters. Embedding-based document-level evaluation for African languages remains unavailable [Q50, Q96, Q175, Q176]. Greedy decoding is used [Q155]. Given the user's MODERATE priority on OF and the partial match (text-only ✓; structural fidelity gap; no plain-language accessibility scoring for variable-literacy end-users), score is 3.

ID	Evidence
Q6	“Furthermore, our analysis reveals that some LLMs exhibit issues such as under-generation, over-generation, repetition of words and phrases, and off-target translations, specifically for translation into African languages.” (p.1)
Q8	“We evaluate performance using automatic metrics and compare the results of encoder-decoder models with decoder-only LLMs” (p.1)
Q44	“Given that our created dataset can be used for sentence-level translation and as a baseline for document-level translation, we evaluate all models on the test splits for each domain.” (p.5)
Q50	“Evaluating document-level translation remains challenging, as existing automatic metrics struggle to capture improvements and account for discourse phenomena (Jiang et al., 2022; Dahan et al., 2024), and embedding-based metrics have not been explored in this context for African languages due to the lack of data.” (p.5)
Q51	“Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF).” (p.5)
Q57	“In Tables 5 and 6 we present d-chrF scores based on the realigned documents, created by merging the translated sentences into their corresponding documents.” (p.5)
Q77	“However, for Yorùbá, it often uses inconsistent or partial diacritics, resulting in inaccuracies.” (p.8)
Q96	“Quality evaluation in MT is an open and ongoing area of research, especially for document-level translation. Recent works have proposed embedding-based metrics for evaluation at both the sentence and document levels. While this has been well explored for high-resource language pairs, it remains underexplored for African languages, although there is a tool, AfriCOMET, that works for sentence-level evaluation in African languages.” (p.10)
Q119	“Please do not delete double empty rows, as they serve to separate paragraphs. Also, avoid deleting any rows, columns, or provided text.” (p.18)
Q155	“In all cases, greedy decoding was used.” (p.21)
Q175	“For evaluation, we used BLEU and chrF scores but excluded AfriCOMET due to its backbone model, AfroXLM-R-L (Alabi et al., 2022; Adelani et al., 2024a), having a context length of 512 tokens.” (p.22)
Q176	“This made it impractical to compute COMET scores for document-level outputs.” (p.22)
Q201	“Given that AFRIDOC-MT is a document-level translation dataset, and due to the limited context length of most translation models and LLMs, which makes it impossible to translate a full document at once, we opted to translate the sentences within the documents and then merge them back to form the complete document. For document-level evaluation, we split the documents into chunks of 10 sentences and translate these chunks using the different models.” (p.24)

Strengths

- Output modality (text in Ethiopic script) directly matches deployment text-only requirement [Q8]
- Paragraph structure preserved at corpus level via empty-row convention [Q119]
- Sentence-to-document realignment supports document-level evaluation [Q51, Q57]
- Output failure modes (under/over-generation, repetition, off-target, diacritic inconsistency) [Q6, Q77] are documented, enabling targeted detection in deployment

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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text output modality matches deployment [Q8, Q44]. Both benchmark and deployment produce Amharic text translations.

OF-2: TTS not assessed by the benchmark. The deployment context indicates extension workers read documents aloud to less-literate cooperative members [regional context: oral_tradition_and_document_mediation], creating an implicit TTS-relevant accessibility requirement that AFRIDOC-MT does not address. INSUFFICIENT DOCUMENTATION on whether deployment system itself includes TTS.

OF-3: Literacy considerations: Ethiopia adult literacy is 51.8% (2017) [WEB-16, WEB-17]; Amhara-region-specific and rural-Amhara-specific rates are not publicly available [regional context: NOT FOUND]. The benchmark does not assess accessibility or comprehension of translated output for variable-literacy readers; reading-level / plain-language scoring is absent. End-users are mediated by cooperative leaders and extension workers, partially mitigating the literacy gap.

OF-4: Form mismatches harming external validity: (a) no structural-fidelity scoring for numbered clauses/tables/signature blocks [Q119 partial preservation only]; (b) no plain-language or readability scoring for variable-literacy readers; (c) documented LLM output pathologies (repetition, off-target) [Q6] would be especially harmful for legal terminology stability in deployment.

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WEB-16	Ethiopia adult literacy 51.8% (2017) — variable-literacy end-users not addressed by benchmark scoring (https://www.theglobaleconomy.com/Ethiopia/Literacy_rate/)
WEB-17	World Bank confirms Ethiopia literacy figure (https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ET)

Information Gaps

- Amhara-region and rural-Amhara literacy rates
- Whether deployment system includes TTS or remains text-only
- Whether deployment delivery format requires preservation of numbered-clause / table / signature-block structure

Requires Expert Verification

- Cooperative leader and extension worker assessment of translation comprehensibility for end-user mediation

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Remediation

Gap 1: No structural-fidelity scoring for numbered clauses, tables, or signature blocks; no readability/accessibility scoring for variable-literacy end-users mediated by cooperative leaders and extension workers.

Recommendation: If deployment delivers structured documents, add a structural-fidelity check (template-preservation rate). Pilot a comprehension test with cooperative leaders and extension workers from one or two woredas (where conflict access permits [WEB-4, WEB-21]) to assess plain-language accessibility of translated output, especially for read-aloud mediation.

ID	Evidence
WEB-4	https://www.gov.uk/government/publications/ethiopia-country-policy-and-information-notes/country-policy-and-information-note-amhara-and-amhara-opposition-groups-ethiopia-june-2025-accessible
WEB-21	https://www.unocha.org/publications/report/ethiopia/ethiopia-situation-report-13-december-2024